STARWEAVER®

Hands-On-Learning: Deception Operations with Cowrie Honeypots

Learning Objectives

By the end of this activity, learners will be able to:

- Configure a basic honeypot to simulate a vulnerable network service.
- Analyze captured log data to track attacker behavior and commands.
- Integrate cyber deception into a broader Defense-in-Depth architecture.



Description

In this hands-on activity, learners will move from theoretical defense to active deception. Using a virtualized Linux environment, learners will interact with a Cowrie honeypot, simulate an attack against it, and analyze the resulting log files to understand how deception tools provide actionable threat intelligence and waste attacker resources.

Structure of the Activity:

Title: Deception Operations with Cowrie Honeypots

Tools used: Virtual Linux Environment, Cowrie, Command Line Terminal

Scenario

You are a Security Engineer managing a Defense-in-Depth architecture. To detect lateral movement and gather threat intelligence, you have decided to deploy a Cowrie honeypot disguised as a vulnerable SSH server. Your task is to simulate a break-in to verify the trap works and then analyze what data the honeypot successfully captured.

Objective

Your goal is to experience how cyber deception tools function as both a detection mechanism and a deterrent. This exercise will help you operate a honeypot in a controlled environment, interpret the data it captures from a simulated attacker session, and connect those findings to concrete defensive actions within a Defense-in-Depth architecture.



Instructions for learners

1. **Exploration & Execution:**

- Access the virtual environment.
- Launch the pre-configured Cowrie honeypot.
- Open a second terminal and simulate an SSH brute-force login using provided "dummy" credentials.
- Once authenticated into the honeypot environment, execute three basic Linux commands: ls, whoami, and cat /etc/passwd.
- Observe how Cowrie accepts and silently records each command without exposing the deception to the simulated attacker.

2 Log Analysis:

- Exit the simulated attack.
- Locate the log entries showing the exact timestamp of the connection, the source IP address used, and each of the three commands executed.
- Note any additional attacker fingerprinting data captured by Cowrie, such as the client version string or attempted credential pairs.

3 Reflection:

- Write down how capturing this specific information would help your organization proactively block future attacks from this adversary.
- Identify one concrete way to integrate this intelligence into your existing firewall rules or threat feed.
- Explain how this honeypot deployment acts as a deterrent by increasing attacker uncertainty and wasting their reconnaissance time.

Organize Information Following This Structure for Peer Review:

Title Page: Activity title, your name, date.

Deception Execution Summary: Describe the steps taken to simulate the attack and the commands executed.

Log Analysis Report: Copy and paste the relevant snippet of the Cowrie log showing the captured "honeytoken" interaction.

Reflections and Future Applications: Explain how this honeypot acts as a deterrent and how the gathered intelligence can be integrated into your firewall rules.

Instructions for Documenting and Sharing the Project for Peer Review

Document the Project

- Use word processing software to create your project summary and reflections.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

Share the Project

- Save your report in PDF format.
- Upload your documents to the “Peer Review” area in the course shell.
- Provide a brief description of your project in the submission post.

Peer Review

- Review the projects submitted by your peers.
- Provide constructive feedback and comments on their log analysis and intelligence application.

This activity gives learners a chance to apply the lesson by operating a real deception tool, interpreting live attacker data, and translating those findings into concrete defensive actions within a layered security architecture.